

Day - 5 MongoDB Assignment

1.show databases

```
test> show dbs  
admin   40.00 KiB  
config 108.00 KiB  
local   64.00 KiB
```

2. create / switch database

```
test> use InsuranceDB  
switched to db InsuranceDB
```

3. create collections

```
InsuranceDB> db.createCollection("Customers")  
{ ok: 1 }  
InsuranceDB> db.createCollection("Policies")  
{ ok: 1 }  
InsuranceDB> db.createCollection("Agents")  
{ ok: 1 }  
InsuranceDB> db.createCollection("Claims")  
{ ok: 1 }  
InsuranceDB> db.createCollection("PolicyAssignments")  
{ ok: 1 }
```

4. Insert documents into Customers

```
db.Customers.insertMany([  
  {  
    CustomerID: 1,  
    FirstName: "Himapriya",  
    LastName: "Pothuganti",  
    DateOfBirth: new Date("2005-03-20"),  
    Phone: "8333807888",
```

```

    Email: "himapriya09@gmail.com"
  },
  {
    CustomerID: 2,
    FirstName: "Aarav",
    LastName: "Sharma",
    DateOfBirth: new Date("1998-07-12"),
    Phone: "9876543210",
    Email: "aarav@gmail.com"
  }
])

{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69575fd791b36ba7ede7a3cb'),
    '1': ObjectId('69575fd791b36ba7ede7a3cc')
  }
}

```

5. Insert documents into Policies

```

db.Policies.insertMany([
  { PolicyID: 101, PolicyName: "Life Secure", PolicyType: "Life", PremiumAmount: 50000,
    DurationYears: 20 },
  { PolicyID: 102, PolicyName: "Health Plus", PolicyType: "Health", PremiumAmount: 30000,
    DurationYears: 10 }
])

{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6957610091b36ba7ede7a3cd'),
    '1': ObjectId('6957610091b36ba7ede7a3ce')
  }
}

```

```
}  
}
```

6. Insert documents into Agents

```
db.Agents.insertMany([  
  { AgentID: 201, AgentName: "Ravi Kumar", Phone: "9123456789", City: "Hyderabad" },  
  { AgentID: 202, AgentName: "Neha Singh", Phone: "9012345678", City: "Bangalore" }  
])
```

```
{  
  acknowledged: true,  
  insertedIds: {  
    '0': ObjectId('6957610091b36ba7ede7a3cd'),  
    '1': ObjectId('6957610091b36ba7ede7a3ce')  
  }  
}
```

7. Insert documents into PolicyAssignments

```
db.PolicyAssignments.insertMany([  
  { AssignmentID: 301, CustomerID: 1, PolicyID: 101, AgentID: 201, StartDate: new Date("2023-01-01"), EndDate: new Date("2043-01-01") },  
  { AssignmentID: 302, CustomerID: 2, PolicyID: 102, AgentID: 202, StartDate: new Date("2022-06-15"), EndDate: new Date("2032-06-15") }  
])
```

```
{  
  acknowledged: true,  
  insertedIds: {  
    '0': ObjectId('6957619f91b36ba7ede7a3d1'),  
    '1': ObjectId('6957619f91b36ba7ede7a3d2')  
  }  
}
```

8. Insert documents into Claims

```
db.Claims.insertMany([  
  { ClaimID: 401, AssignmentID: 301, ClaimDate: new Date("2024-05-10"), ClaimAmount: 15000,  
    ClaimStatus: "Approved" },  
  { ClaimID: 402, AssignmentID: 302, ClaimDate: new Date("2023-11-20"), ClaimAmount: 8000,  
    ClaimStatus: "Pending" }  
])
```

```
{  
  acknowledged: true,  
  insertedIds: {  
    '0': ObjectId('695761ab91b36ba7ede7a3d3'),  
    '1': ObjectId('695761ab91b36ba7ede7a3d4')  
  }  
}
```

9. Display one customer document.

```
InsuranceDB> db.Customers.findOne()  
  
{  
  _id: ObjectId('69575fd791b36ba7ede7a3cb'),  
  CustomerID: 1,  
  FirstName: 'Himapriya',  
  LastName: 'Pothuganti',  
  DateOfBirth: ISODate('2005-03-20T00:00:00.000Z'),  
  Phone: '8333807888',  
  Email: 'himapriya09@gmail.com'  
}
```

10. Display agent whose AgentID is 202

```
InsuranceDB> db.Agents.find({AgentID:202})  
  
[  
  {
```

```
_id: ObjectId('6957619691b36ba7ede7a3d0'),
AgentID: 202,
AgentName: 'Neha Singh',
Phone: '9012345678',
City: 'Bangalore'
}
]
```

11. Display only PolicyName and PremiumAmount (using projection)

```
InsuranceDB> db.Policies.find(
  {},
  { PolicyName: 1, PremiumAmount: 1, _id: 0 }
)
[
  { PolicyName: 'Life Secure', PremiumAmount: 50000 },
  { PolicyName: 'Health Plus', PremiumAmount: 30000 }
]
```

12. Update city of agent with AgentID = 201

```
InsuranceDB> db.Agents.updateOne(
  {AgentID:201},
  {$set :{City:'Chennai'}}
)
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

13. Decrease ClaimAmount of all pending claims by 1000

```
InsuranceDB> db.Claims.updateMany(
  { ClaimStatus: "Pending" },
  { $inc: { ClaimAmount: -1000 } }
)
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

14. Delete one agent whose AgentID is 202

```
InsuranceDB> db.Agents.deleteOne({ AgentID: 202 })
{ acknowledged: true, deletedCount: 1 }
```

15. Delete all claims whose ClaimAmount is less than 10000

```
InsuranceDB> db.Claims.deleteMany({ ClaimAmount: { $lt: 10000 } })
{ acknowledged: true, deletedCount: 1 }
```

16. Find customer where CustomerID equals 1

```
InsuranceDB> db.Customers.find({ CustomerID: { $eq: 1 } })
[
  {
    _id: ObjectId('69575fd791b36ba7ede7a3cb'),
    CustomerID: 1,
    FirstName: 'Himapriya',
    LastName: 'Pothuganti',
    DateOfBirth: ISODate('2005-03-20T00:00:00.000Z'),
    Phone: '8333807888',
    Email: 'himapriya09@gmail.com'
  }
]
```

```
}  
]
```

16. Find customers named Himapriya with Gmail account

```
InsuranceDB> db.Customers.find({  
... $and: [  
... { FirstName: "Himapriya" },  
... { Email: { $regex: "gmail" } }  
... ]  
... })  
[  
  {  
    _id: ObjectId('69575fd791b36ba7ede7a3cb'),  
    CustomerID: 1,  
    FirstName: 'Himapriya',  
    LastName: 'Pothuganti',  
    DateOfBirth: ISODate('2005-03-20T00:00:00.000Z'),  
    Phone: '8333807888',  
    Email: 'himapriya09@gmail.com'  
  }  
]
```

17. Find policies of type Life or Health

```
InsuranceDB> db.Policies.find({ PolicyType: { $in: ["Life", "Health"] } })  
[  
  {  
    _id: ObjectId('6957610091b36ba7ede7a3cd'),  
    PolicyID: 101,  
    PolicyName: 'Life Secure',  
    PolicyType: 'Life',  
    PremiumAmount: 50000,
```

```
    DurationYears: 20
  },
  {
    _id: ObjectId('6957610091b36ba7ede7a3ce'),
    PolicyID: 102,
    PolicyName: 'Health Plus',
    PolicyType: 'Health',
    PremiumAmount: 30000,
    DurationYears: 10
  }
]
```

18. Display customers sorted by DateOfBirth

```
InsuranceDB> db.Customers.find().sort({ DateOfBirth: 1 })
```

```
[
  {
    _id: ObjectId('69575fd791b36ba7ede7a3cc'),
    CustomerID: 2,
    FirstName: 'Aarav',
    LastName: 'Sharma',
    DateOfBirth: ISODate('1998-07-12T00:00:00.000Z'),
    Phone: '9876543210',
    Email: 'aarav@gmail.com'
  },
  {
    _id: ObjectId('69575fd791b36ba7ede7a3cb'),
    CustomerID: 1,
    FirstName: 'Himapriya',
    LastName: 'Pothuganti',
    DateOfBirth: ISODate('2005-03-20T00:00:00.000Z'),
    Phone: '8333807888',
    Email: 'himapriya09@gmail.com'
  }
]
```

```
}  
]
```

19. Display top 1 most expensive policy

```
[  
  {  
    _id: ObjectId('6957610091b36ba7ede7a3cd'),  
    PolicyID: 101,  
    PolicyName: 'Life Secure',  
    PolicyType: 'Life',  
    PremiumAmount: 50000,  
    DurationYears: 20  
  }  
]
```

20. Display total number of customers

```
InsuranceDB> db.Customers.countDocuments()  
  
2
```

21. Display distinct cities of agents

```
InsuranceDB> db.Agents.distinct("City")  
  
[ 'Chennai' ]
```

22. Display claims raised in year 2024

```
InsuranceDB> db.Claims.find({  
  ClaimDate: {  
    $gte: new Date("2024-01-01"),  
    $lte: new Date("2024-12-31")  
  }  
})  
  
[  
  {
```

```

    _id: ObjectId('695761ab91b36ba7ede7a3d3'),
    ClaimID: 401,
    AssignmentID: 301,
    ClaimDate: ISODate('2024-05-10T00:00:00.000Z'),
    ClaimAmount: 15000,
    ClaimStatus: 'Approved'
  }
]

```

23. Display customers whose email starts with 'h'.

```

InsuranceDB> db.Customers.find({ Email: { $regex: "^h" } })

[
  {
    _id: ObjectId('69575fd791b36ba7ede7a3cb'),
    CustomerID: 1,
    FirstName: 'Himapriya',
    LastName: 'Pothuganti',
    DateOfBirth: ISODate('2005-03-20T00:00:00.000Z'),
    Phone: '8333807888',
    Email: 'himapriya09@gmail.com'
  }
]

```

24. Join Customers with PolicyAssignments (lookup)

```

InsuranceDB> db.Customers.aggregate([
  {
    $lookup: {
      from: "PolicyAssignments",
      localField: "CustomerID",
      foreignField: "CustomerID",
      as: "Policies"
    }
  }
])

```

```
  ])  
  [  
    {  
      _id: ObjectId('69575fd791b36ba7ede7a3cb'),  
      CustomerID: 1,  
      FirstName: 'Himapriya',  
      LastName: 'Pothuganti',  
      DateOfBirth: ISODate('2005-03-20T00:00:00.000Z'),  
      Phone: '8333807888',  
      Email: 'himapriya09@gmail.com',  
      Policies: [  
        {  
          _id: ObjectId('6957619f91b36ba7ede7a3d1'),  
          AssignmentID: 301,  
          CustomerID: 1,  
          PolicyID: 101,  
          AgentID: 201,  
          StartDate: ISODate('2023-01-01T00:00:00.000Z'),  
          EndDate: ISODate('2043-01-01T00:00:00.000Z')  
        }  
      ]  
    },  
    {  
      _id: ObjectId('69575fd791b36ba7ede7a3cc'),  
      CustomerID: 2,  
      FirstName: 'Aarav',  
      LastName: 'Sharma',  
      DateOfBirth: ISODate('1998-07-12T00:00:00.000Z'),  
      Phone: '9876543210',  
      Email: 'aarav@gmail.com',  
      Policies: [  

```

```

{
  _id: ObjectId('6957619f91b36ba7ede7a3d2'),
  AssignmentID: 302,
  CustomerID: 2,
  PolicyID: 102,
  AgentID: 202,
  StartDate: ISODate('2022-06-15T00:00:00.000Z'),
  EndDate: ISODate('2032-06-15T00:00:00.000Z')
}
]
}
]

```

25. Display total claim amount

```

db.Claims.aggregate([
  { $group: { _id: null, TotalClaimAmount: { $sum: "$ClaimAmount" } } }
])

[ { _id: null, TotalClaimAmount: 15000 } ]

```

26. Display total claim amount per customer

```

db.PolicyAssignments.aggregate([
  {
    $lookup: {
      from: "Claims",
      localField: "AssignmentID",
      foreignField: "AssignmentID",
      as: "Claims"
    }
  },
  { $unwind: "$Claims" },
  {
    $group: {

```

```
    _id: "$CustomerID",
    TotalClaim: { $sum: "$Claims.ClaimAmount" }
  }
}
])
[ { _id: 1, TotalClaim: 15000 } ]
```

27. Create index on CustomerID

```
InsuranceDB> db.Customers.createIndex({ CustomerID: 1 })
CustomerID_1
```

28. Find customers who do NOT have any policies

```
InsuranceDB> db.Customers.aggregate([
  { $lookup: {
    from: "PolicyAssignments",
    localField: "CustomerID",
    foreignField: "CustomerID",
    as: "policies"
  }},
  { $match: { policies: { $size: 0 } } }
])
```